

Game GIRL - Engineering Math & Formula Reference

Game GIRL V1

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Scope

This document is the working mathematics reference for the **Game GIRL V1** architecture:

- Raspberry Pi 4 Model B
- STM32U575 / NUCLEO-U575ZI-Q
- 1S LiPo battery
- 5 V boost converter
- MAX17043 fuel gauge
- Elecrow 3.5 inch 480 x 320 ILI9488 SPI LCD
- MAX98357A audio amplifier
- physical buttons and optional rumble

It collects the formulas likely to be useful for electronics, battery sizing, charging, wiring, GPIO, I2C, timers, debouncing, display bandwidth, audio, storage, emulator timing, performance, and thermal checks.

1. SI prefixes

$$\begin{aligned}k &= 10^3 \\M &= 10^6 \\G &= 10^9 \\m &= 10^{-3} \\\mu &= 10^{-6} \\n &= 10^{-9} \\p &= 10^{-12}\end{aligned}$$

Examples:

- $5 \text{ mA} = 0.005 \text{ A}$
- $100 \text{ }\mu\text{s} = 0.0001 \text{ s}$
- $1 \text{ MHz} = 1,000,000 \text{ Hz}$

2. Ohm's law

$$V = IR$$

$$I = \frac{V}{R}$$

$$R = \frac{V}{I}$$

Where V is voltage in volts, I current in amperes, and R resistance in ohms.

Example, 3.3 V across 330 ohms:

$$I = \frac{3.3}{330} = 0.010 \text{ A} = 10 \text{ mA}$$

3. Electrical power

$$P = VI$$

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

Example:

$$5 \text{ V} \times 4 \text{ A} = 20 \text{ W}$$

4. Total current for parallel loads

For devices connected in parallel to the same rail:

$$I_{total} = I_1 + I_2 + I_3 + \dots + I_n$$

Recommended supply margin:

$$I_{rated} \geq 1.2 \text{ to } 1.3 \times I_{peak}$$

Example: if the estimated peak is 3.4 A, with 25% margin:

$$3.4 \times 1.25 = 4.25 \text{ A}$$

5. Energy

$$E = Pt$$

Battery energy in watt-hours:

$$E_{Wh} = V_{nominal} C_{Ah}$$

For a 3.7 V, 5000 mAh battery:

$$5000 \text{ mAh} = 5 \text{ Ah}$$

$$E = 3.7 \times 5 = 18.5 \text{ Wh}$$

6. Battery runtime

First approximation:

$$t = \frac{E_{battery} \eta}{P_{load}}$$

Where η is converter efficiency as a decimal.

For an 18.5 Wh battery, 90% efficiency and 10 W average load:

$$t = \frac{18.5 \times 0.9}{10} \approx 1.67 \text{ h}$$

At 15 W:

$$t \approx 1.11 \text{ h}$$

At 20 W:

$$t \approx 0.83 \text{ h}$$

7. Average load power

If power changes over time:

$$P_{avg} = \frac{P_1 t_1 + P_2 t_2 + \dots + P_n t_n}{t_1 + t_2 + \dots + t_n}$$

Example, 5 W for 30 min and 12 W for 30 min:

$$P_{avg} = \frac{5(0.5) + 12(0.5)}{1} = 8.5 \text{ W}$$

8. Battery discharge C-rate

$$I_{max} = C_{rate} C_{Ah}$$

For a 5 Ah cell rated 2C:

$$I_{max} = 2 \times 5 = 10 \text{ A}$$

9. Boost converter efficiency

$$\eta = \frac{P_{out}}{P_{in}}$$

Therefore:

$$P_{in} = \frac{P_{out}}{\eta}$$

For 20 W output at 90% efficiency:

$$P_{in} = \frac{20}{0.90} \approx 22.22 \text{ W}$$

10. Boost converter battery current

This is one of the most important formulas for Game GIRL:

$$I_{in} = \frac{V_{out} I_{out}}{V_{in} \eta}$$

Example, 5 V at 4 A from a 3.7 V battery at 90% efficiency:

$$I_{in} = \frac{5 \times 4}{3.7 \times 0.90} \approx 6.0 \text{ A}$$

Near low battery voltage, 3.0 V:

$$I_{in} = \frac{20}{3.0 \times 0.90} \approx 7.4 \text{ A}$$

11. Boost converter loss

$$P_{loss} = P_{in} - P_{out}$$

Equivalent form:

$$P_{loss} = P_{out} \left(\frac{1}{\eta} - 1 \right)$$

For 20 W output at 90% efficiency:

$$P_{loss} \approx 2.22 \text{ W}$$

12. Ideal boost converter duty cycle

For an ideal boost converter:

$$D = 1 - \frac{V_{in}}{V_{out}}$$

At 3.7 V to 5 V:

$$D = 1 - \frac{3.7}{5} \approx 0.26$$

At 3.0 V to 5 V:

$$D = 1 - \frac{3.0}{5} = 0.40$$

13. Charge-time estimate

Ideal estimate:

$$t_{charge} \approx \frac{C_{Ah}}{I_{charge}}$$

For 5 Ah at 2 A:

$$t = \frac{5}{2} = 2.5 \text{ h}$$

Because Li-ion charging includes a constant-voltage phase, a practical estimate is:

$$t_{real} \approx 1.1 \text{ to } 1.3 \times \frac{C}{I}$$

14. Charge C-rate

$$C_{charge} = \frac{I_{charge}}{C_{Ah}}$$

For a 5 Ah cell charged at 2 A:

$$C_{charge} = \frac{2}{5} = 0.4C$$

15. Wire voltage drop

$$V_{drop} = IR$$

Example: 50 milliohms total path resistance at 6 A:

$$V_{drop} = 6 \times 0.05 = 0.30 \text{ V}$$

16. Wire power loss

$$P_{wire} = I^2 R$$

At 6 A through 50 milliohms:

$$P = 6^2 \times 0.05 = 1.8 \text{ W}$$

17. Conductor resistance

$$R = \rho \frac{L}{A}$$

For copper:

$$\rho \approx 1.68 \times 10^{-8} \Omega m$$

Remember that the round-trip current path includes both outgoing and return conductors.

18. LED series resistor

$$R = \frac{V_{GPIO} - V_F}{I_{LED}}$$

Example, 3.3 V GPIO, 2.0 V LED drop, 5 mA current:

$$R = \frac{3.3 - 2.0}{0.005} = 260 \Omega$$

Choose the next common value such as 270 ohms.

19. Voltage divider

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$

Example with 100k/100k and 4.2 V input:

$$V_{out} = 4.2 \times \frac{100}{200} = 2.1 \text{ V}$$

20. ADC conversion

For an N -bit ADC:

$$ADC_{max} = 2^N - 1$$

Input voltage:

$$V_{in} = \frac{ADC_{code}}{2^N - 1} V_{ref}$$

For a 12-bit ADC, 2048 code and 3.3 V reference:

$$V \approx \frac{2048}{4095} \times 3.3 \approx 1.65 \text{ V}$$

21. Pull-up current

When an open-drain I2C line is low:

$$I = \frac{V}{R_P}$$

For 3.3 V and 4.7 kohms:

$$I \approx 0.70 \text{ mA}$$

22. I2C rise time

Approximation:

$$t_r \approx 0.8473 R_P C_B$$

Therefore:

$$R_P \leq \frac{t_r}{0.8473 C_B}$$

For 200 pF total bus capacitance and 300 ns maximum rise time:

$$R_P \lesssim 1.77 \text{ k}\Omega$$

23. RC time constant

$$\tau = RC$$

After one time constant the capacitor reaches about 63.2% of final voltage. After about five time constants it is above 99%.

24. RC low-pass cutoff

$$f_c = \frac{1}{2\pi RC}$$

For 10 kohms and 100 nF:

$$f_c \approx 159 \text{ Hz}$$

25. Software debounce sample count

If buttons are sampled every T_s :

$$N = \left\lceil \frac{t_{debounce}}{T_s} \right\rceil$$

For 10 ms debounce and 1 ms sampling:

$$N = 10$$

26. Button bit masks

For button index n :

$$mask = 1 \ll n$$

Example in C:

```
BTN_A = 1u << 0;  
BTN_B = 1u << 1;  
BTN_L = 1u << 9;
```

27. STM32 timer frequency

For a common STM32 timer configuration:

$$f_{update} = \frac{f_{timer}}{(PSC + 1)(ARR + 1)}$$

Period:

$$T = \frac{1}{f_{update}}$$

Example, 160 MHz timer clock, PSC=159, ARR=999:

$$f = \frac{160,000,000}{160 \times 1000} = 1000 \text{ Hz}$$

which gives a 1 ms period.

28. PWM duty cycle

$$D \approx \frac{CCR}{ARR + 1}$$

Percentage:

$$D_{\%} = 100 \frac{CCR}{ARR + 1}$$

29. Frequency and period

$$f = \frac{1}{T}$$

$$T = \frac{1}{f}$$

Examples:

- 1 kHz -> 1 ms
- 60 Hz -> 16.67 ms
- 48 kHz -> 20.83 us/sample

30. Display pixel count

$$N_{pixels} = WH$$

For the Game GIRL LCD:

$$480 \times 320 = 153600 \text{ pixels}$$

For the GBA framebuffer:

$$240 \times 160 = 38400 \text{ pixels}$$

31. Integer scaling

$$S_x = \frac{W_{display}}{W_{source}}$$

$$S_y = \frac{H_{display}}{H_{source}}$$

For Game GIRL:

$$S_x = S_y = 2$$

32. Aspect ratio

$$AR = \frac{W}{H}$$

GBA:

$$\frac{240}{160} = 1.5 = 3 : 2$$

Game GIRL display:

$$\frac{480}{320} = 1.5 = 3 : 2$$

33. Framebuffer memory

$$M = WHB_{pp}$$

For RGB565, $B_{pp} = 2$ bytes/pixel.

480 x 320:

$$480 \times 320 \times 2 = 307200 \text{ bytes}$$

about 300 KiB.

GBA 240 x 160:

$$240 \times 160 \times 2 = 76800 \text{ bytes}$$

about 75 KiB.

34. Double-buffer memory

$$M_{double} = 2M_{frame}$$

For 480 x 320 RGB565:

$$614400 \text{ bytes}$$

about 600 KiB.

35. Raw display bandwidth

$$BW = WHB_{pp}FPS$$

For 480 x 320, RGB565, 60 FPS:

$$BW = 480 \times 320 \times 2 \times 60$$

$$BW = 18,432,000 \text{ bytes/s} \approx 18.4 \text{ MB/s}$$

This excludes protocol overhead.

36. SPI transfer time

$$t = \frac{N_{bits}}{f_{SPI}}$$

or:

$$t = \frac{N_{bytes} \times 8}{f_{SPI}}$$

For one 307,200-byte frame at 40 MHz:

$$t = 61.44 \text{ ms}$$

Ideal maximum full-frame rate:

$$FPS \approx 16.3$$

At 80 MHz:

$$t = 30.72 \text{ ms}, \quad FPS \approx 32.6$$

37. ILI9488 3-byte pixel mode

If the SPI path uses 3 bytes/pixel:

$$M = 480 \times 320 \times 3 = 460800 \text{ bytes/frame}$$

At 40 MHz:

$$t = 92.16 \text{ ms}$$

$$FPS \approx 10.85$$

At 80 MHz:

$$t = 46.08 \text{ ms}$$

$$FPS \approx 21.7$$

This is important for the Elecrow ILI9488 module and must be measured in practice.

38. Dirty-rectangle bandwidth

If only fraction k of the screen changes:

$$BW_{partial} = kBW_{full}$$

At 25% update area:

$$BW_{partial} = 0.25 \times 18.4 \approx 4.6 \text{ MB/s}$$

39. SPI utilization

Theoretical byte bandwidth:

$$BW_{SPI} = \frac{f_{SPI}}{8}$$

At 40 MHz:

$$BW_{SPI} = 5 \text{ MB/s}$$

Utilization:

$$U = \frac{BW_{required}}{BW_{SPI}}$$

If $U > 1$, the requested transfer rate is physically impossible.

40. Pixels per inch

$$PPI = \frac{\sqrt{W^2 + H^2}}{D}$$

For 480 x 320 at 3.5 inches:

$$PPI \approx 165$$

41. Audio data rate

$$R = f_s N_{channels} \frac{\text{bits}}{8}$$

At 48 kHz, 16-bit mono:

$$R = 48000 \times 1 \times 2 = 96000 \text{ bytes/s}$$

Stereo:

$$R = 192000 \text{ bytes/s}$$

42. Audio sample period

$$T_s = \frac{1}{f_s}$$

At 48 kHz:

$$T_s \approx 20.83 \mu s$$

43. Audio buffer duration

$$t_{buffer} = \frac{N_{samples}}{f_s}$$

For 1024 samples at 48 kHz:

$$t \approx 21.33 \text{ ms}$$

44. Speaker RMS current

For a simplified resistive load:

$$I_{RMS} = \sqrt{\frac{P}{R}}$$

For 3 W into 4 ohms:

$$I_{RMS} \approx 0.866 \text{ A}$$

45. Speaker RMS voltage

$$V_{RMS} = \sqrt{PR}$$

For 3 W into 4 ohms:

$$V_{RMS} \approx 3.46 \text{ V}$$

46. Storage transfer time

$$t = \frac{S}{R}$$

Example, 32 MiB ROM at effective 20 MiB/s:

$$t \approx 1.6 \text{ s}$$

47. Binary memory units

$$1KiB = 1024B$$

$$1MiB = 1024KiB$$

$$1GiB = 1024MiB$$

A 32 MiB ROM contains:

$$32 \times 1024 \times 1024 = 33,554,432 \text{ bytes}$$

48. GBA master clock

The GBA CPU clock is approximately:

$$f_{GBA} = 16,777,216 \text{ Hz}$$

Guest time for N cycles:

$$t = \frac{N}{16,777,216}$$

49. GBA scanline timing

A GBA scanline lasts 1232 cycles and there are 228 scanlines per frame:

$$cycles_{frame} = 1232 \times 228 = 280,896$$

50. GBA frame time

$$t_{frame} = \frac{280896}{16777216} \approx 16.74 \text{ ms}$$

51. GBA frame rate

$$FPS = \frac{16,777,216}{280,896} \approx 59.73$$

This is the target, not exactly 60 Hz.

52. Emulator cycle ratio

A rough host-to-guest clock ratio is:

$$R = \frac{f_h}{f_{guest}}$$

This is only a theoretical upper bound. Real emulator work includes instruction decoding, memory dispatch, timing, PPU, audio, interrupts, and branches.

53. Emulator real-time condition

For real-time emulation:

$$t_{emulate \ frame} \leq t_{GBA \ frame}$$

Therefore:

$$t_{emulate \ frame} \leq 16.74 \text{ ms}$$

If one frame takes 20 ms:

$$FPS_{effective} = \frac{1}{0.020} = 50 \text{ FPS}$$

54. Emulator speed percentage

$$Speed_{\%} = 100 \frac{t_{target}}{t_{actual}}$$

For target 16.74 ms and actual 20 ms:

$$Speed \approx 83.7\%$$

55. Audio samples per GBA frame

$$samples/frame = \frac{f_s}{FPS_{GBA}}$$

At 48 kHz:

$$\frac{48000}{59.73} \approx 803.6$$

Because this is not an integer, use a fractional accumulator to prevent long-term drift.

56. Fractional accumulator

Generic pattern:

```
accumulator += numerator

while accumulator >= denominator:
    accumulator -= denominator
    emit_event()
```

This is useful for audio/video synchronization and other rate conversions.

57. Fixed-point representation

For n fractional bits:

$$x_{real} = \frac{x_{integer}}{2^n}$$

Real to fixed:

$$x_{fixed} = round(x_{real}2^n)$$

Multiplication:

$$c = \frac{ab}{2^n}$$

Typical C implementation:

```
int32_t c = (int32_t)((int64_t)a * b) >> n);
```

58. Linear interpolation

$$y = a + t(b - a)$$

with:

$$0 \leq t \leq 1$$

Useful for audio resampling and UI animation.

59. Percentage

$$percentage = 100 \frac{part}{whole}$$

CPU utilization:

$$CPU_{\%} = 100 \frac{busy\ time}{total\ time}$$

60. Frame-time profiling

Average frame time:

$$\bar{t} = \frac{\sum_{i=1}^N t_i}{N}$$

FPS:

$$FPS = \frac{1}{\bar{t}}$$

If time is in milliseconds:

$$FPS = \frac{1000}{t_{ms}}$$

61. Moving average

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_N}{N}$$

Useful for telemetry and performance measurements.

62. Exponential moving average

$$y_n = \alpha x_n + (1 - \alpha)y_{n-1}$$

Smaller α gives stronger smoothing.

63. Percentage error

$$Error_{\%} = 100 \frac{|measured - expected|}{|expected|}$$

64. Frequency error in ppm

$$Error_{ppm} = 10^6 \frac{f_{measured} - f_{target}}{f_{target}}$$

65. Thermal rise approximation

If thermal resistance is known:

$$\Delta T = P_{loss} \theta$$

Junction temperature estimate:

$$T_J = T_A + P_{loss} \theta_{JA}$$

This is only a first-order estimate for complete modules.

66. Measured power-stage efficiency

$$\eta = \frac{V_{out} I_{out}}{V_{in} I_{in}}$$

Example:

- $V_{in} = 3.7$ V
- $I_{in} = 5.8$ A
- $V_{out} = 5.0$ V
- $I_{out} = 3.8$ A

$$\eta \approx 88.5\%$$

67. Power-budget formula

For each device:

$$P_i = V_i I_i$$

Total:

$$P_{total} = \sum P_i$$

With design margin M :

$$P_{design} = P_{total} M$$

Typical initial margin:

$$M = 1.2 \text{ to } 1.3$$

68. Runtime from measured current

Quick estimate:

$$t_{hours} \approx \frac{C_{Ah}}{I_{avg}}$$

For 5 Ah and 4 A average battery current:

$$t \approx 1.25 h$$

Energy-based runtime is usually better when a boost converter is involved.

69. Remaining energy from state of charge

$$E_{remaining} \approx E_{full} \times SOC$$

For 18.5 Wh full and 60% SOC:

$$E_{remaining} \approx 11.1 Wh$$

70. Estimated time remaining

$$t_{remaining} \approx \frac{E_{remaining}}{P_{avg}}$$

Use averaged power rather than instantaneous power to avoid a jumpy estimate.

71. USB HID report bandwidth

$$BW = Sf$$

For an 8-byte report at 1000 reports/s:

$$BW = 8000 \text{ bytes/s}$$

This is tiny compared with USB bandwidth.

72. Sampling frequency requirement

Nyquist condition:

$$f_s > 2f_{max}$$

In practice, use substantial margin so analog filtering remains realistic.

73. Aliasing

A useful alias relation is:

$$f_{alias} = |f - kf_s|$$

for integer k chosen so the alias falls in the observable band.

74. Decibels - voltage ratio

For equal impedances:

$$dB = 20 \log_{10} \left(\frac{V_2}{V_1} \right)$$

75. Decibels - power ratio

$$dB = 10 \log_{10} \left(\frac{P_2}{P_1} \right)$$

76. One-pole digital smoothing

$$y_n = y_{n-1} + \alpha(x_n - y_{n-1})$$

Equivalent to the exponential moving average.

77. Memory alignment

Round x upward to alignment a , where a is a power of two:

$$aligned = (x + a - 1) \& \sim (a - 1)$$

C macro:

```
#define ALIGN_UP(x, a) (((x) + (a) - 1) & ~((a) - 1))
```

Useful for DMA and cache-line alignment.

78. Ring-buffer occupancy

For a circular buffer of size N :

```
used = (write_index - read_index + N) % N;  
free = N - used - 1;
```

79. Circular index increment

General case:

$$index_{next} = (index + 1) \bmod N$$

If N is a power of two:

```
index = (index + 1) & (N - 1);
```

80. Battery UI smoothing

The MAX17043 already estimates state of charge. For display-only smoothing:

$$SOC_{display} = \alpha SOC_{new} + (1 - \alpha) SOC_{old}$$

Do not replace the gauge algorithm with a simple voltage-to-percentage conversion.

81. Pre-power calculations

Main 5 V rail

$$P_{5V} = 5V \times I_{total}$$

Target design capability is approximately 4 A or more.

Battery current

$$I_{battery} = \frac{P_{5V}}{V_{battery}\eta}$$

Evaluate at 4.2 V, 3.7 V, and 3.0 V. The worst case is normally near low battery voltage.

Battery capability

$$I_{battery,max} < C_{rate}C_{Ah}$$

The rated battery discharge current should exceed the calculated peak with margin.

Wire drop

$$V_{drop} = IR$$

Converter heat

$$P_{loss} = P_{out} \left(\frac{1}{\eta} - 1 \right)$$

82. Display feasibility calculation

Before relying on the SPI screen:

$$bytes/frame = width \times height \times bytes/pixel$$

$$transfer\ time = \frac{bytes/frame \times 8}{SPI\ clock}$$

$$max\ ideal\ FPS = \frac{1}{transfer\ time}$$

If ideal FPS is already below the target, software optimization cannot overcome the raw link limit.

83. Emulator timing calculations

Use:

$$cycles/frame = 280896$$

$$frame\ time \approx 16.74\ ms$$

$$FPS \approx 59.73$$

Advance CPU, timers, DMA, PPU, audio and interrupts using emulated GBA cycles rather than host wall-clock guesses.

84. Useful constants

Quantity	Value
GBA CPU clock	16,777,216 Hz
GBA cycles per scanline	1,232
GBA scanlines per frame	228
GBA cycles per frame	280,896
GBA frame rate	about 59.73 Hz
GBA framebuffer	240 x 160
Game GIRL LCD	480 x 320
Integer scale	2x
RGB565	2 bytes/pixel
LiPo nominal voltage	3.7 V
LiPo full voltage	4.2 V
Main system rail	5.0 V
Rough boost efficiency assumption	0.90
Target 5 V current capability	about 4 A or more

85. Recommended measurement log

During prototype testing, record:

- battery voltage;
- battery current;
- 5 V output voltage;
- 5 V output current;
- boost converter temperature;
- Raspberry Pi temperature;
- LCD current;
- idle power;
- emulator power;
- Moonlight power if used;
- audio maximum-volume power;
- battery SOC;
- frame time;
- display update time;
- audio buffer underruns.

Then calculate:

$$P = VI$$

$$\eta = \frac{P_{out}}{P_{in}}$$

$$runtime \approx \frac{E_{battery}\eta}{P_{avg}}$$

86. Quick-reference formula sheet

$$V = IR$$

$$P = VI$$

$$Wh = V \times Ah$$

$$t \approx \frac{Wh \times \eta}{P_{avg}}$$

$$I_{in} = \frac{V_{out} I_{out}}{V_{in} \eta}$$

$$\eta = \frac{P_{out}}{P_{in}}$$

$$V_{drop} = IR$$

$$P_{wire} = I^2 R$$

$$R_{LED} = \frac{V_s - V_F}{I_{LED}}$$

$$V_{out} = V_{in} \frac{R_2}{R_1 + R_2}$$

$$V_{ADC} = \frac{code}{2^N - 1} V_{ref}$$

$$\tau = RC$$

$$f_c = \frac{1}{2\pi RC}$$

$$f_{timer} = \frac{f_{clk}}{(PSC + 1)(ARR + 1)}$$

$$BW = WHB_{pp}FPS$$

$$t_{SPI} = \frac{bytes \times 8}{f_{SPI}}$$

$$Audio\ bytes/s = f_s \times channels \times bytes/sample$$

$$t_{GBA} = \frac{cycles}{16,777,216}$$

$$\Delta T = P\theta$$

87. Engineering principle

For every subsystem:

calculate



add margin



measure



compare with calculation



adjust design

The formulas give the first design. The multimeter, oscilloscope, logic analyzer, profiler and thermometer determine whether the real Game GIRL behaves as expected.